A General-Purpose Computing Core With Cooperative Motion Detection and Feature Extraction for Always-On PWM Image Sensors

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***Abstract*—This article presents a mixed-signal general-purpose computing core designed to support real-time inference applications in always-on low-power pulsewidth modulation (PWM) CMOS image sensors (CISs), functioning as a processing-in-sensor (PIS) circuit. This core can be integrated into the columns of CISs to perform low-power edge processing on images, without affecting the pixel fill factor and imaging quality. It employs a coordinated mechanism in which motion detection (MD) triggers the activation of feature extraction (FE), thereby achieving an organic integration of MD and FE functionalities, and maximizing the system's power efficiency. MD is implemented via in-column frame difference (FD), while FE is performed using a programmable-weight 3×3 convolution, a rectified linear unit activation function, and a 2×2 max-pooling (MP) operation. Both functionalities are computed based on real-time PWM signals from the CIS and the principle of current integration, with partial circuit reuse achieved through different switching operations. A 0.8-V computing core prototype, with an area of 720×272 μm, was fabricated and verified using 0.18-μm standard CMOS technology. The experimental results at an image frame rate of 250 fps demonstrated an average power consumption of 5.23 μW for MD and 17.53 μW for FE. The prototype core computes the first two layers of an ultra lightweight convolutional neural network (CNN) for the task of MNIST digit classification, achieving an accuracy loss of only 0.86% compared to the ideal scenario. This analog computing core can be used in multimode, low-power, edge-intelligent vision sensors.**

***Index Terms*—CMOS image sensor (CIS), convolutional neural network (CNN), feature extraction (FE), general-purpose computing core, motion detection (MD), processing in sensor (PIS).**

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I. Introduction

With the vigorous development of machine vision applications such as intelligent transportation and security surveillance systems, the realization of an always-on edge-intelligent CMOS image sensor (CIS) with low power consumption and high energy efficiency has become increasingly urgent. In recent years, significant progress has been achieved in low-power image capture schemes based on pulsewidth modulation (PWM) readout technology [1], [2], [3], [4], with power consumption reduced to microwatt level, representing a reduction of an order of magnitude compared to a traditional voltage readout CIS. Moreover, this approach eliminates the dynamic range (DR) limitations associated with the supply voltage [5]. However, in a conventional image processing architecture that combines CISs with dedicated AI accelerators, the transfer of raw image data between the imager and the processor incurs even more substantial power consumption [6], [7]. This fundamental limitation of conventional architectures prevents edge devices from achieving always-on, long-term operation under stringent power budgets.

To overcome the above issues, CISs with processing-in-sensor (PIS) technology have been developed [8], [9], [10], [11], [12], [20], [26]. In this technology, image data preprocessing is performed directly within the sensor, effectively eliminating substantial redundant information. Only the extracted key feature information is transferred to off-chip processors, which carry out computationally intensive operations for subsequent high-precision computing. As a result, this significantly reduces the data throughput of the ADC, thereby substantially lowering the overall system power consumption. This is because the quantization operation of the ADC is responsible for the majority (>50%) of the total power consumption of a conventional CIS [13]. However, continuously performing feature extraction (FE) on static images still leads to significant power and bandwidth inefficiencies in always-on edge devices. Hence, motion detection (MD) is performed on the image before FE, to eliminate static images and trigger the FE operation only when moving objects are detected. This MD triggers the FE mechanism, which ensures

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minimal processing of redundant image data while ensuring complete moving target retention.

Previous authors have attempted to integrate MD with region of interest (ROI) extraction methods [14], [15]; however, an ROI is limited to extracting localized regions within an image and therefore fails to capture the comprehensive contextual information of the entire image. In addition, some works have combined MD with the histogram-of-oriented-gradient (HOG) algorithm [16], although these have still relied on traditional active pixel structures and the integration of ADC with digital SRAM, resulting in high overall power consumption.

Furthermore, frame memory is essential for MD. Consequently, for in-sensor MD, frame difference (FD)-based vision sensors typically employ in-pixel analog storage capacitors and processing circuits [17], [18], [19], which inevitably lead to highly complex pixel structures, low fill factors, and poor imaging quality. Researchers into in-sensor FE operations have proposed an in-column computing architecture based on a tiny convolutional neural network (CNN) [20]. However, the essential comparator module in this architecture still incurs significant power consumption.

To summarize, the prevailing challenges in PIS systems predominantly manifest in three critical aspects.

1) The presence of redundant image information leads to power wastage during data transmission and processing.
2) An FD scheme based on the in-pixel architecture suffers from the issue of a low pixel fill factor.
3) The comparator module in the in-column architecture has a high power consumption.

This article therefore proposes, for the first time, a general-purpose computing core that is suitable for all PWM CISs. This core can be deployed within the CIS column, directly receiving the PWM signals output from the pixel array and performing computations in the mixed-signal domain. It has the following characteristics.

1) By implementing a coordination mechanism that triggers FE based on MD, an organic integration between FD-based MD and tiny-CNN-based FE is achieved. This approach maximizes the elimination of redundant image information, resulting in energy-efficient image processing with ultralow power consumption.
2) The leveraging of a novel codesign methodology that achieves deep hardware integration between MD and FE operations through optimized circuit reuse.
3) The incorporation of a dynamic tail current source comparator achieves an optimized balance between performance and power consumption.

This innovative architecture fundamentally differs from conventional functional module stacking approaches by employing a wake-up mechanism to achieve system-level power savings. Through optimized circuit reuse, the design maintains multifunctional capabilities while effectively addressing area overhead constraints. Furthermore, departing from traditional in-pixel MD architectures, the proposed strategy eliminates the need for additional pixel circuitry, thereby preserving high fill factor while ensuring compatibility with diverse PWM CIS configurations.

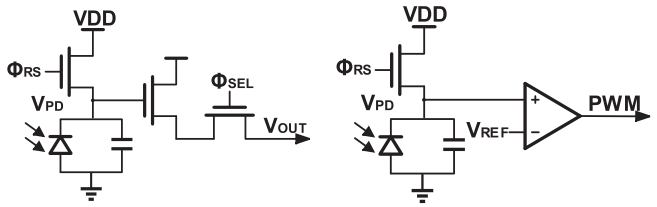


Fig. 1. Schematics of pixel structures (a) traditional APS pixel structure and (b) PWM pixel structure.

The remainder of this article is organized as follows. Section II discusses related PWM CISs. Section III describes the proposed in-column mixed-signal computing core for CIS with a low-power dynamic tail current source comparator. Section IV presents the experimental results for the prototype CIS chip, and Sections V and VI provide a conclusion and a discussion, respectively.

II. Related PWM CISs

Fig. 1(a) and (b) shows the schematics of the traditional active pixel sensor (APS) and the PWM pixel structures, respectively. In comparison to the conventional APS pixel, the primary distinction of the PWM pixel lies in its integration of an in-pixel comparator. This comparator is capable of converting analog voltage signals into time-domain PWM signals. The design of the in-pixel comparator encompasses various implementation methodologies, each with unique features in performance, power consumption, and area efficiency.

Chiou and Hsieh [2] proposed a PWM CIS with threshold variation cancellation. To reduce the pixel area and improve the fill factor, their pixel architecture incorporates a common-source amplifier composed of two transistors, which functions as a comparator to determine the edge transition timing of the generated PWM signal, thereby producing a PWM signal that is linearly proportional to the photocurrent. Kim et al. [5] proposed a CIS with a token readout and repeated-scanning scheme. The pixel circuit was designed with a two-stage dynamic comparator and a token latch, in which the output of the comparator controlled the mode of the token latch and the accessibility of the column line to the token latch.

Although the pixel structures and array scales of such PWM CISs vary, the computing core proposed in this study is fully compatible with their in-column architectures. By deploying a corresponding number of computing cores, efficient PIS functionality can be achieved, which illustrates the architectural versatility and system adaptability of the proposed core. The integration of a low-power PWM CIS with the proposed general-purpose in-sensor computing core enables the realization of low-power CISs with edge-intelligent processing capabilities. This architecture achieves a 60% improvement over the conventional CIS processing solutions [7] in energy efficiency, providing an efficient hardware solution for the development of intelligent vision systems.

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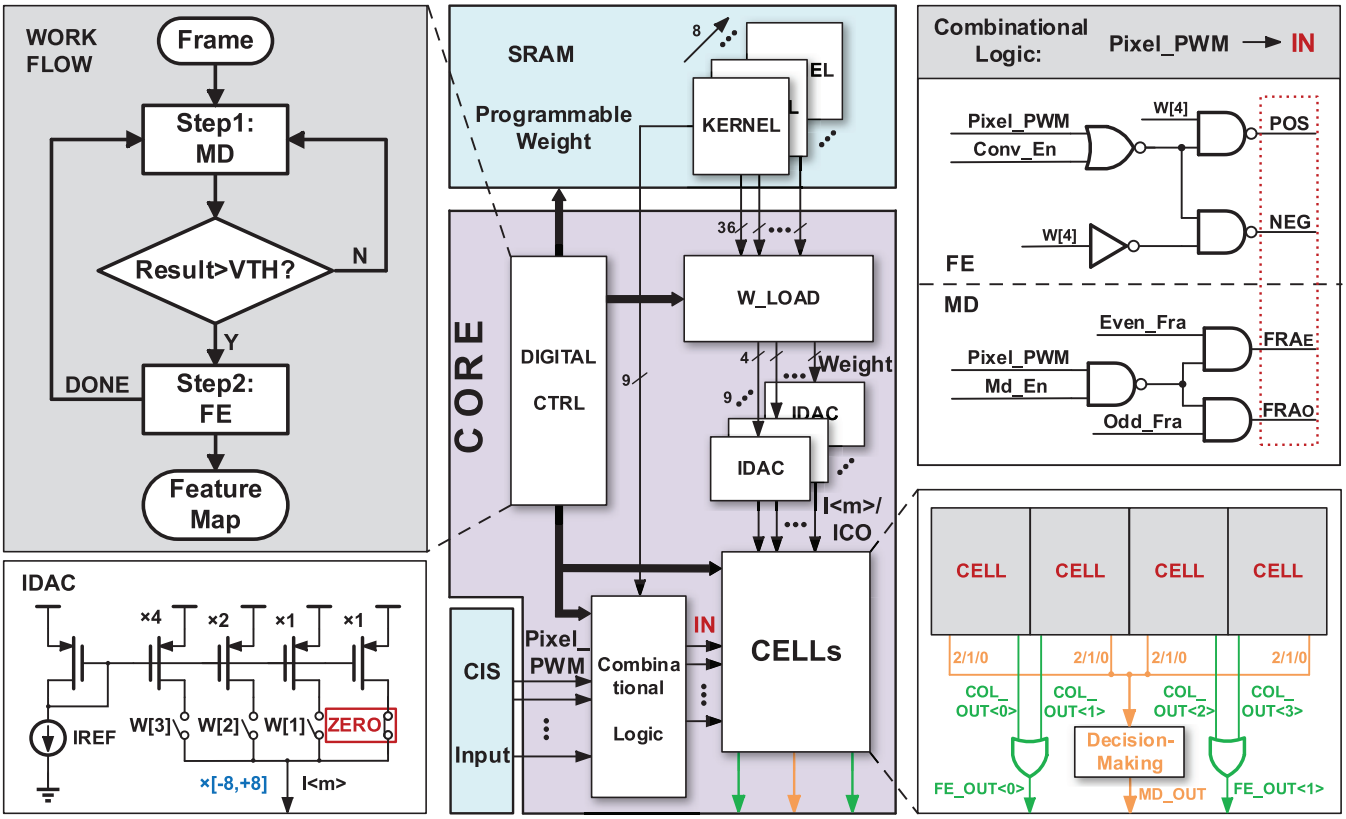


Fig. 2. Block diagram of the proposed computing core.

III. Proposed Computing Core

This section provides a comprehensive description of the proposed general-purpose computing core, including a detailed analysis of the structure and functionalities. In addition, an overall scheme for integrating the computing core with CISs is presented.

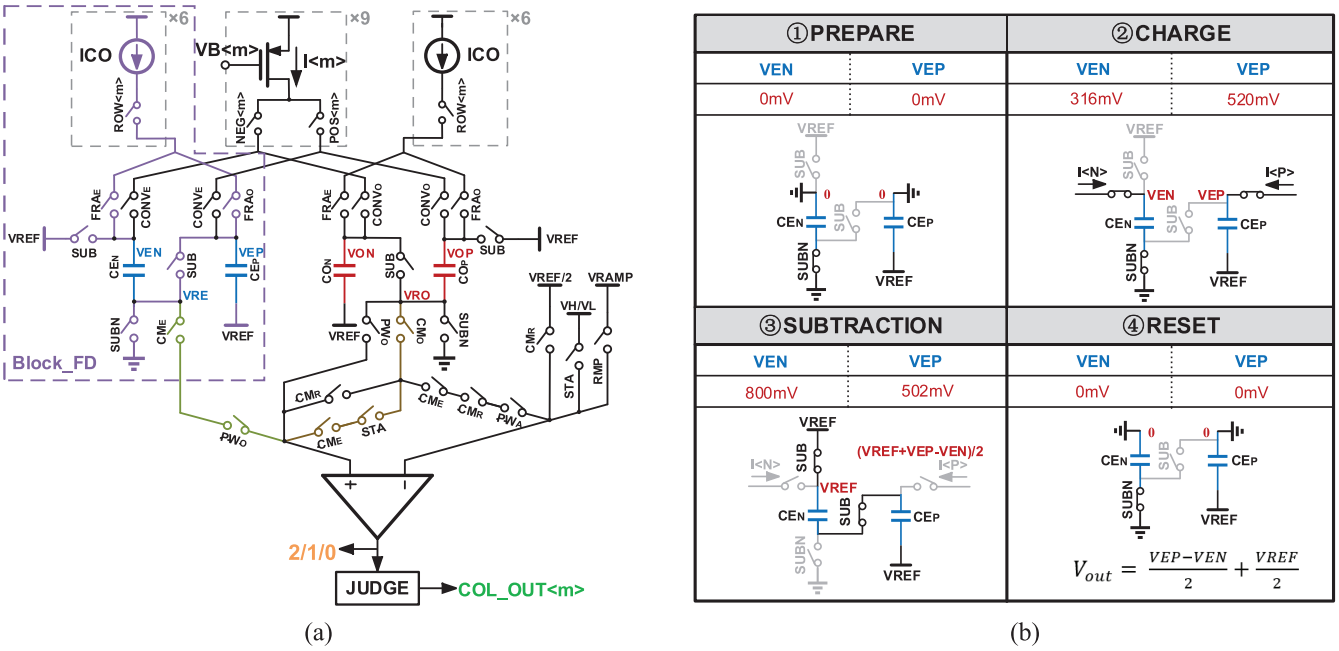
A. Overall Architecture

Fig. 2 illustrates the overall architecture of the proposed computing core, which consists of four computing cells, a digital control module, a weight loading module, IDACs for digital-to-current conversion, a decision-making module, and simple digital logic gates. The digital control module employs a robust finite-state-machine architecture for mode transition. During the MD mode, FD is performed exclusively on a designated subset of pixel regions, while the output data from the final scanlines are intentionally discarded, thereby ingeniously repurposing this temporal interval as a critical decision-making window. State transitions are strictly synchronized to the frame-synchronized clock's rising edge, enabling seamless FE mode activation at the next frame boundary when motion is detected, or maintaining low-power MD mode otherwise. This design achieves dual advantages: significantly reduced power consumption in the MD mode while eliminating the inter-frame data hazards, thereby making the computing core particularly suitable for practical implementations. Additionally, upon MD, the system executes a mandatory five-frame FE processing sequence before returning to MD mode, ensuring comprehensive feature capture. The computing core executes either serial FD operations for MD or fully parallel multiply-accumulate (MAC) operations for FE through digital timing control in different modes, thereby accomplishing a two-stage process for image information. In MD mode, the computing core generates an MD_OUT signal which is counted by a counter, whereas in the FE mode, it produces two parallel FE_OUT signals.

To create the input to the computing core, PWM signals are converted into switch control signals (POS, NEG, FRA_E, and FRA_O) through simple logic gate combinations, which clamp the current corresponding to different modes. To create the weights to the computing core, in the IDAC module, the digital weight signals ($W[3], W[2], W[1]$, and ZERO) are converted into proportional currents by controlling the switch states of the current-mirror branches. By incorporating the ZERO signal, the weight range can be represented as $[-8, +8]$ when considering the signs of the weights. This range ensures adequate feature learning capability [20] while maintaining hardware costs within practically acceptable limits for edge devices. Since the MD operation does not require weights, only the ZERO switch needs to be closed to obtain a constant current for computing.

The core of the convolution operation—MAC—is implemented through a mixed-domain method based on weighted current integration [21], [22], expressed as IT = CV. The PWM signal controls the switches to clamp the charging time

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(a) (b)

Fig. 3. Proposed computing cell (a) architecture of the cell and (b) subtraction computing process.

and polarity of the current to the capacitor, thus transforming the multiplication of two multibit input values into the integration of current over constant capacitors in the mixed-signal domain. As a result, the integrated charge is proportional to the product of the input and the weight, which can be expressed as follows:

$$Q_s = \sum_i |I(x) \times T(x)| = C_P \times V_P + C_N \times V_N. \quad (1)$$

The current integration-based MAC operation demonstrates significant advantages by overcoming the fundamental limitations inherent in conventional analog computing architectures. Specifically, while voltage-domain architectures achieve high operational speed, they suffer from substantial power overhead due to the mandatory analog-to-digital/digital-to-analog (AD/DA) conversions [23]. Phase-domain approaches incur excessive power consumption caused by the high-frequency oscillation requirements of digitally controlled oscillators (DCOs) [24]. Although time-domain architectures also employ PWM for signal conversion, their multiplication mechanism based on simple AND gates necessitates multiple delay chains and digital adders to support multibit operations [25]. Consequently, the current integration approach maintains computational accuracy while effectively circumventing the power consumption and complexity bottlenecks inherent in the conventional architectures.

B. Proposed General-Purpose Computing Core

Fig. 3(a) illustrates the architecture of the proposed computing cell, which consists of current-mirror transistors, four capacitors, a dynamic tail current comparator, and several switches. Since each cell contains four capacitors, it is capable of storing two sets of frame information and performing subtraction operations between them. The subtraction operation is achieved by connecting the plates of two capacitors [26]. Taking capacitors CE_N and CE_P as an example, the switch operation steps are shown in Fig. 3(b). Initially, the lower plates of capacitors CE_N and CE_P are connected to GND and VREF, respectively, while the upper plates are grounded and reset to 0 V. During the current integration phase, charge accumulates on capacitors CE_N or CE_P, causing the upper plate voltages to rise to VEP and VEN, respectively, while the lower plates remain connected to GND and VREF. When the current integration is complete, the switches controlled by the SUB signal are closed, while the switches controlled by the SUBN signal are opened, causing the upper plate of CE_N to connect to VREF and the lower plate to switch from GND to the upper plate of CE_P. As a result, the upper plate voltage of CE_N becomes VREF, while the lower plate voltage, which should theoretically become VREF−VEN due to the disconnection from GND, is redistributed due to its connection to the upper plate of CE_P, resulting in a voltage of ((VREF − VEN + VEP)/2). Thus, the expression for the output voltage includes the difference between the voltages of the two capacitors and a reference voltage. VREF was set to 600 mV to ensure both computing accuracy and energy efficiency.

C. In-Column FD for MD

The proposed computing core performs MD by executing FD, following the operational flowchart and timing diagram illustrated in Fig. 4(a) and (b). ICO represents the constant current generated by the bandgap reference current and the current mirror a negative feedback loop. ROW⟨m⟩ denotes the row control signal for the selected the pixel block performing FD, where each pixel block is composed of a 6×1 pixel aggregation. The aim of this process is to reduce detection blind spots and increase the detectable pixel coverage by 6×,

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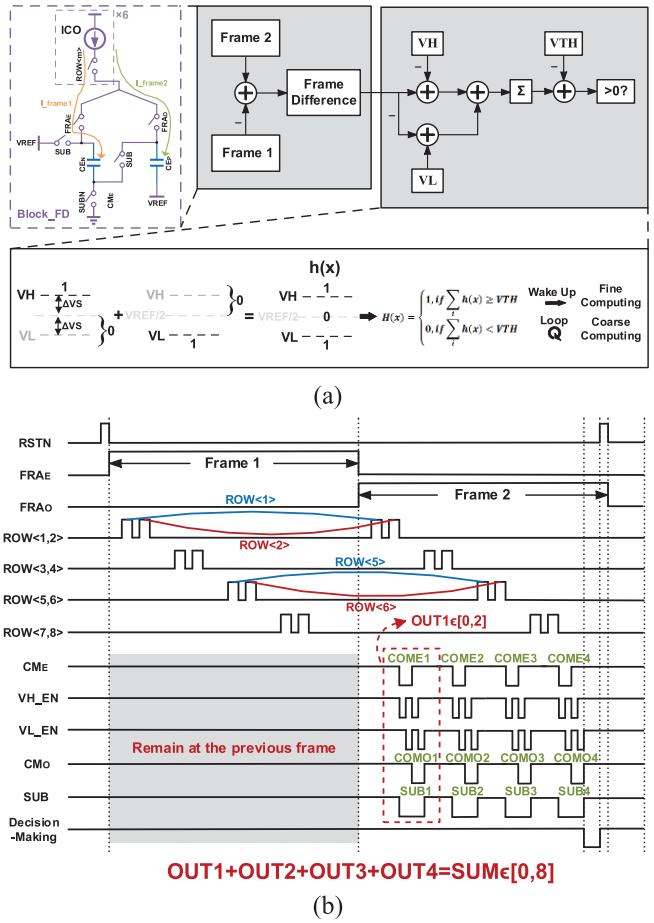


Fig. 4. FD operation (a) flowchart and (b) timing diagram.

thereby preventing small objects from escaping detection [27]. FRA_E and FRA_O represent the effective times of the input signals within even and odd frames, respectively, and are derived from the input signal Pixel_PWM through combinational logic gates. Six sets of constant currents are accumulated onto two capacitors, CE_N and CE_P, through two switch controls, and the current paths in the previous frame (frame 1) and the current frame (frame 2) are indicated in orange and green, respectively. The subtraction is then completed by connecting the capacitor plates. CO_N and CO_P follow the same process, resulting in the following FD expression:

$$V_{FD} = V_{CR} - V_{PR} + \delta \tag{2}$$

where V_{CR} and V_{PR} represent the accumulation of information on the capacitors for the pixel blocks performing MD in frames 2 and 1, respectively, and δ represents the error due to charge leakage [28]. In general, the higher the frame rate, the smaller the error. Subsequently, the STA switch closes, and one end of the comparator is connected to the reference voltage, while the other end is connected to two branches [denoted in green and brown in Fig. 3(a)], whose switches close sequentially. The two voltages are then compared with VH and VL in turn, in order to determine whether the two pixel blocks contain moving objects.

It is important to note that in digital circuits, the absolute value of the subtraction result must be taken before comparing it with the threshold. Hence, in the analog circuit, we set thresholds voltages VH and VL symmetrically around the reference voltage (VREF/2) and compare the subtraction result with these thresholds sequentially. Here, VH − (VREF/2) = (VREF/2)−VL = ΔVS. The corresponding region of the image is considered to have undergone changes only if the voltage exceeds VH or falls below VL; otherwise, it is deemed to remain static. The result of the first comparison, if greater than VH, is recorded as a value of one, and if less than VH, as zero. The result of the second comparison, if less than VL, is recorded as one, and if greater than VL, as zero. The results of the two comparisons are summed and recorded as $h(x)$. When the voltage values fall between VH and VL, $h(x) = 0$; otherwise, $h(x) = 1$. Thus, the FD signal is quantized to a 1-bit signal. The individual comparisons are performed during the pixel row time intervals in frame 2 that are not selected for MD. Finally, after all the MD pixel blocks have been compared, the results are summed and compared with a preset total threshold VTH. If the sum exceeds VTH, it is concluded that there is an object in motion in the scene, and FE is performed on the next frame; otherwise, the scene is considered to be static, and MD continues. Note that this study employs a static-adaptive VTH configuration strategy to enhance motion target detection sensitivity and environmental disturbance rejection, in accordance with the design objectives for low-power edge applications. The implementation involves a joint optimization process of threshold parameters (VH, VL, and VTH) through iterative MD operations on multiple training image sets prior to scene-specific MD execution, ultimately converging to an optimal threshold configuration.

D. In-Column FE

In contrast to MD, the core aspect of FE lies in replacing the convolutional MAC operations with weighted current integration, rather than employing a constant current for integration. The MAC operation convolves the image with multiple 3×3 kernels using a stride of three, thus requiring nine sets of weighted currents. The pMOS transistor generates the weighted current $I\langle m\rangle$ by replicating the gate voltage $VB\langle m\rangle$. $POS\langle m\rangle$ and $NEG\langle m\rangle$ are obtained by combining the sign bit of the weight $W[4]$ with the PWM signals output from the pixels, through combinational logic. $CONV_E$ and $CONV_O$ represent the effective activation times for even and odd rows of pixels, respectively. Fig. 5(a) illustrates the timing diagram of the FE operation, with its detailed workflow described as follows. The operating sequence of switches involved in the comparison process is shown in Fig. 5(b).

1) During the readout period of three rows of pixels in the even array, the nine sets of weighted currents $I\langle m\rangle$ are multiplied through the clamping of switches $POS\langle m\rangle$ and $NEG\langle m\rangle$, and the results are accumulated and subtracted on capacitors CE_N and CE_P to obtain the intermediate result VRE.
2) Subsequently, the same operations are performed on CO_N and CO_P to derive the intermediate results for the three rows of pixels in the odd array, denoted as VRO.

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3) VRE and VRO are sequentially connected to the input of the comparator, with the other input connected to the reference voltage (VREF/2). This is because the result of subtraction of the capacitor voltages includes an additional bias of (VREF/2). If the computed result is greater than (VREF/2), this indicates a positive result, and the capacitor voltage remains unchanged. If the result is less than (VREF/2), the connected plate is connected to GND for reset, which is equivalent to performing the operation of the nonlinear activation function ReLU.
4) VRE and VRO are connected to the two inputs of the comparator, respectively, to perform 2×1 max pooling (MP). If VRE > VRO, VRE remains connected to the input, while the connected plates generating VRO are reset to GND. If VRE < VRO, VRO is connected to the input via the green path through the PW_O switch, and the connected plates generating VRE are reset to GND. The other input of the comparator is connected to the ramp voltage VRAMP through a switch controlled by the RMP signal, which converts the voltage into a COL_OUT signal.
5) Finally, two adjacent output COL_OUT signals are processed through an OR gate to obtain the maximum value, FE_OUT, thus completing the 2×2 MP operation.

In this way, through multiple iterations, the proposed computing core can perform FE on the entire image, significantly reducing the power consumption associated with data transfer.

E. Dynamic Tail Current Source Comparator

Fig. 6(a) shows a schematic of the proposed dynamic tail current source comparator, which consists of a two-stage dynamic amplifier and a positive feedback loop [29]. Capacitors CL1 and CL2 are loads of the first and second stages, respectively, and during the preparation phase, they are reset by their respective reset switches, M9 and M12. The PUL signal is a pulse signal (set to 100 MHz). Fig. 6(b) shows the timing diagram and operation principle. With VINN held constant, VINP was applied with either a red-annotated ramp voltage or a blue-annotated constant voltage. The simulation results demonstrate that under the modulation of the switching control signal PUL, VC exhibits periodic quasi-conical waveform characteristics. This behavior stems from the comparator's dual-mode operation.

1) *High-Performance Active Phase (PUL = 1):* When the comparator is enabled, the capacitor CCHA acts as a tail current source to generate a significant tail current.
2) *Low-Power Standby Phase (PUL = 0):* VDD charges CCHA through M1, and there is no current path from VDD to GND. Consequently, the node voltage rapidly decays, causing the tail current to drop sharply. Meanwhile, the output voltage VOUT1 remains unchanged due to CL1.

This dynamic tail current regulation mechanism optimizes the average tail current at a minimal level across the full operational period while preserving substantial gain during critical decision-making phases, thereby achieving an optimized tradeoff between power consumption and performance. The differential pair employs M5 and M6 as cross-coupled loads, and to improve the gain and common-mode stability, branches M7 and M8 are incorporated.

The second stage is a simple common-source amplifier, with M10 acting as the amplifying transistor and M11 controlling the enabling of the second stage. The output of the second stage, VOUT2, is connected to a positive feedback loop. When the VINP terminal exceeds VINN, the feedback loop rapidly pulls VOUT2 to GND through M13, thereby reducing the delay of OUT and improving linearity. In addition, it can be regarded as a level shifter, with the output voltage determined by the voltage of the inverter, in order to match the voltage level of the overall circuit. Since the comparison time accounts for only a portion of the whole period, the average power consumption of the comparator is low, at 0.43 μW.

TABLE I
CHIP SPECIFICATIONS

	Specification
Technology	180nm CMOS
Supply voltage	0.8V (digital and analog)
Area	2618 × 1020μm
	473 × 38μm (per cell)
MAC I/O/W precision	8b/8b/ [−8, +8]
Power	MD: 5.23μW (total), 1.31μW (cells)
	FE: 17.53μW (total), 3.78μW (cells)
Energy efficiency (FE)	1.5TOPS/W

IV. EXPERIMENTAL RESULTS

A 0.8-V mixed-signal general-purpose computing core prototype used in multimode always-on CIS was fabricated using 0.18-μm 1P6M standard CMOS technology. Fig. 7 shows a photograph of the proposed computing core test chip on an evaluation board. A computing core with an area of 720×272 μm is integrated onto the test chip. This core comprises four computing units, along with digital peripheral timing control circuits, a current reference, and a current-mirror array. The chip is integrated onto a printed circuit board designed for evaluation and tested through the chip-on-board process. Fig. 8 illustrates the overall experimental environment. The testing setup included the chipboard, power supply, oscilloscope, multimeter, and a field-programmable gate array (FPGA) for controlling test execution and data transfer, along with programming tools for firmware uploading.

Table I summarizes the specifications of the test chip. The supply voltage for both digital and analog components is 0.8 V. The entire test chip area occupies 2618×1020 μm, while each computing cell has a size of 473×38 μm. The inputs and outputs of the MAC computation are PWM signals (quantizable to 8 bits), with weight precision ranging from [−8, +8]. The effective operation periods for the MD and FE modes are 8.064 ms and 24 μs, respectively. The corresponding

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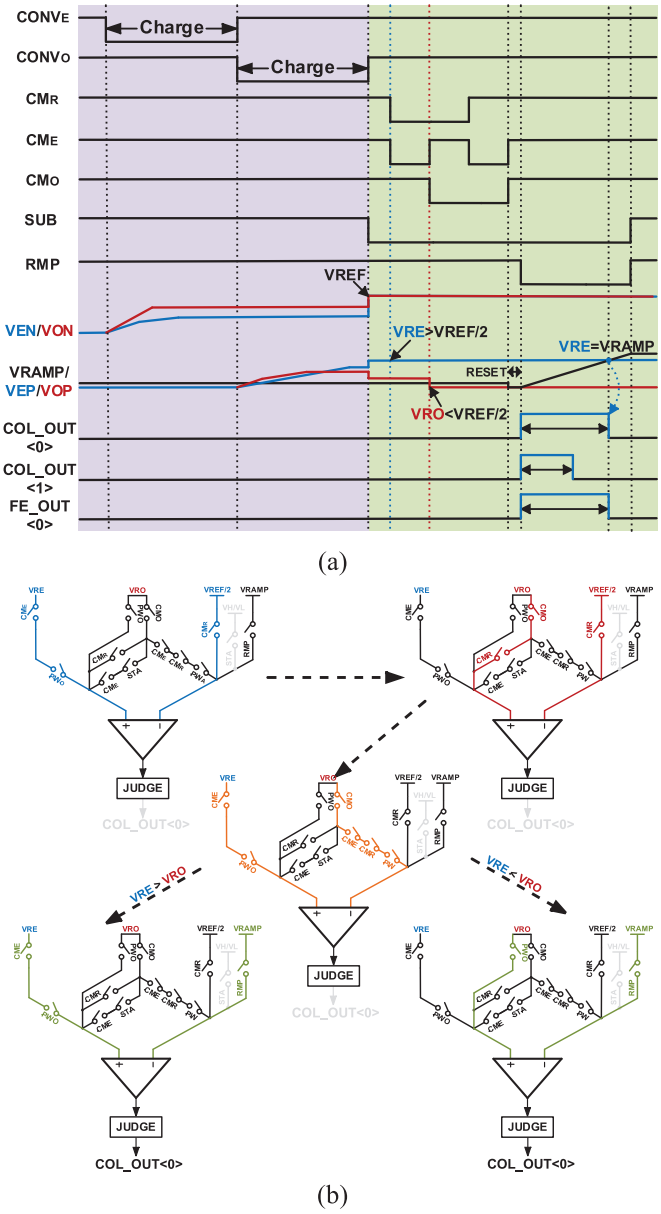


Fig. 5. FE operation (a) timing diagram and (b) operating sequence.

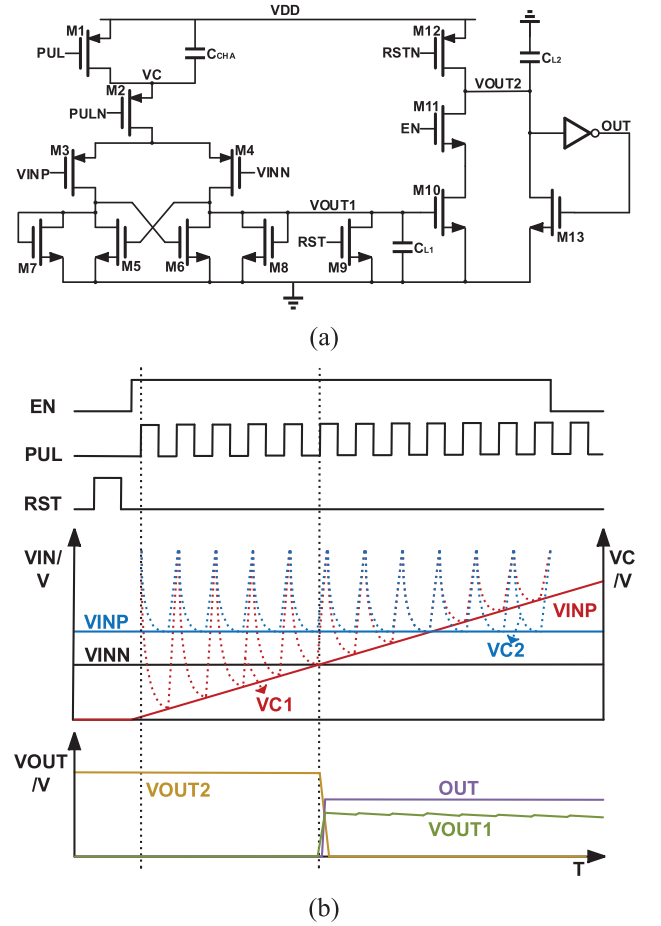


Fig. 6. Dynamic tail current source comparator (a) schematic and (b) timing diagram and operation principle.

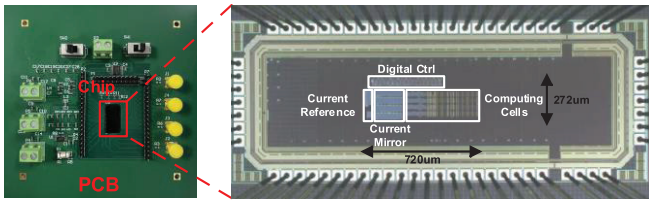


Fig. 7. Overall scheme of the proposed core for CIS.

average power consumption values for the computing core are 5.23 and 17.53 μW, while the average power consumption values for the four computing cells are 1.31 and 3.78 μW, respectively. In the FE mode, the energy efficiency of the computing cell is 1.5 TOPS/W.

We assume that the overall testing architecture is based on a 120×120 pixel array that provides PWM signals to multiple computing cores. However, only the central 63×63 pixels are uniformly distributed sampled for MD. In other words, when an object intrusion is detected in the center of the frame, the MD identifies the moving object and wakes up the FE module to process the entire frame and output the feature map. Otherwise, the system continues to perform MD. To test a single computing core, the test image is first converted into corresponding 8-bit grayscale values using Python commands. These grayscale values are then proportionally transformed into PWM signals, representing the image information, which is input into the chip through the FPGA for computing. When the chip has completed the computing process, the resulting PWM signals are sent back to the FPGA, where they are counted and converted into digital values. These digital values are either summed to generate the MD judgment result or transferred to the PC for further processing, such as conversion into grayscale images or subsequent CNN computations. The entire experimental process is illustrated in Fig. 9.

To evaluate the MD capability of the computing core, a 63×63 resolution image was taken as the reference image at the initial time point, in which a square with adjustable size and velocity served as the moving object. Similar to [27], we varied the size and movement speed of the square to generate post-movement images. Both the initial and

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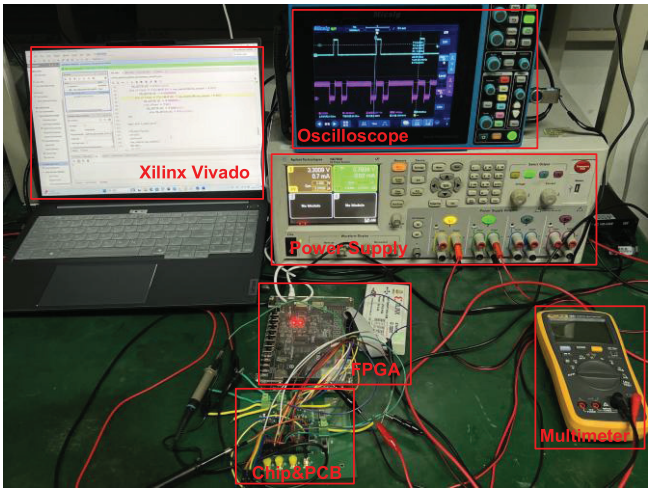


Fig. 8. Experimental environment.

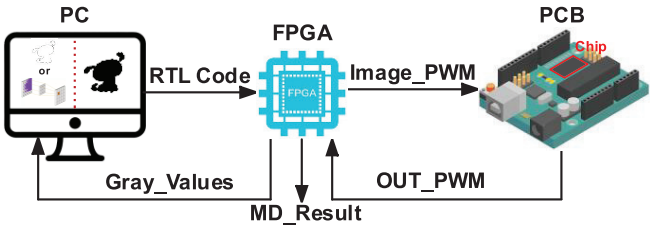


Fig. 9. Entire experimental procedure.

post-movement images were converted into PWM signals and fed into the computing core for FD processing. The FD operates at a frame rate of 250 fps, with the pulsewidth of the PWM signal corresponding to the maximum grayscale value of 255 set to 6.4 μs. The average power consumption of MD is 5.23 μW. The sum of the number of pixel blocks detecting motion events is used as the performance index. For our evaluation case, uniformly distributed sampling was employed, with the MD frame resolution set to $6 \times 8 \times 10$ pixels. As a result, the computing core performed ten consecutive calculations to obtain the FD results, which were then compared with the preset threshold. Due to the effect of the error δ arising from charge leakage, the voltage accumulated on the capacitor during the previous frame experiences greater leakage. Thus, the threshold voltages VH and VL are set to 320 and 290 mV, respectively.

Fig. 10(a) shows the initial image and the post-movement images under different influencing conditions. Fig. 10(b) illustrates the relationship between the total number of pixel blocks detected as motion events and the start-to-end distance of objects of the same size, thereby reflecting the impact of object motion speed on the total number of events. The greater the start-to-end distance, the faster the object's speed of motion; conversely, a smaller distance indicates a slower motion speed. We use uniform distribution sampling with 0° defined as the rightward direction, and the motion of the object is analyzed separately in the 0°, 90°, and 135° directions, as these three cases effectively represent the motion of the

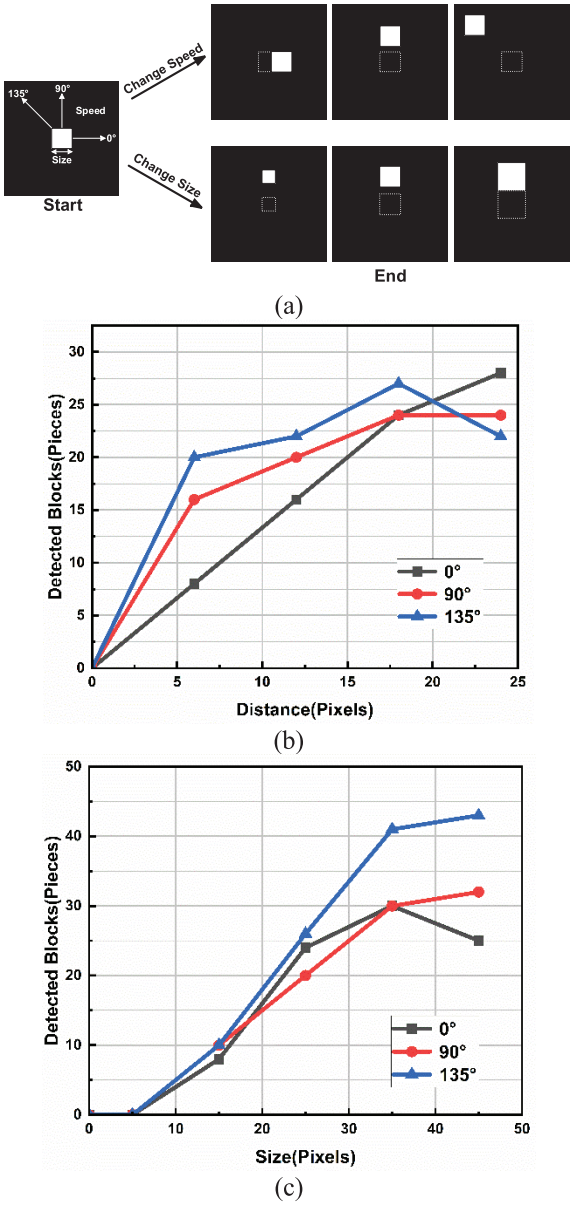


Fig. 10. MD characterization (a) initial and post-movement images, (b) total detected events versus object speed/displacement (object size = 21 pixels), and (c) total detected events versus object size (object distance = 15 pixels).

object in any arbitrary direction. As expected, FD is highly sensitive to variations in the object's position. The trend of the curve shows that the total number of events detected by FD is generally proportional to the object speed, but that the detection performance is limited in the slow-motion region. Reducing the FD frame rate can increase the displacement of the object, thereby improving detection performance in the slow-motion region, albeit at the cost of a slower system response. If the object moves beyond the image frame, the number of detected events undergoes a slowing in the growth trend or even a decrease in all three directions. Fig. 10(c) shows the relationship between the total number of pixel blocks detected as motion events and the object size. As the object size increases, the number of events detected by FD also exhibits an increasing trend. Similarly, if the object is

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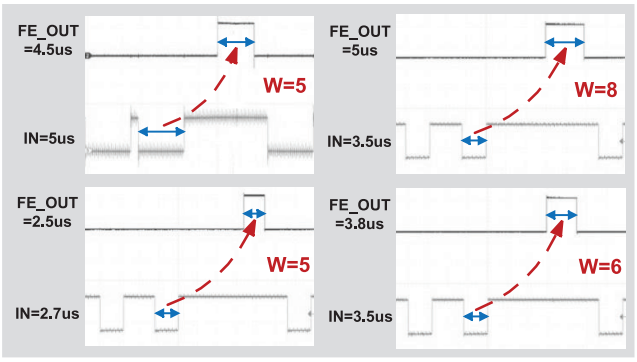


Fig. 11. Experimental results for MAC (a) variation in input and (b) variation in weight.

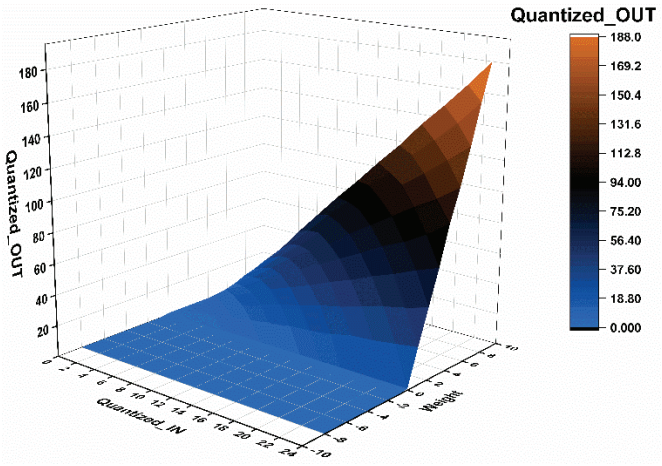


Fig. 12. Basic input-output characteristics for MAC.

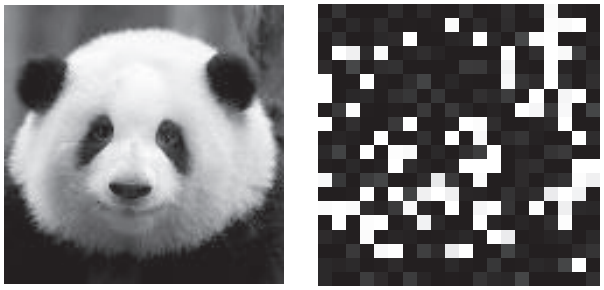


Fig. 13. (a) Captured raw image (120 × 120) and (b) feature map after edge extraction and downsampling (20 × 20).

too large and exceeds the image boundaries, a decrease in the number of detected events may occur. For very small objects, high-frame-rate FD may fail to detect the object, while low-frame-rate FD can mitigate this issue.

To validate FE, it is essential to first verify the accuracy of the MAC operations performed by the computing core. Fig. 11(a) and (b) presents the experimental results for the output PWM signals obtained by varying the input values and the positive weights. The output is proportional to both the input and the positive weights. We normalized the pulsewidth

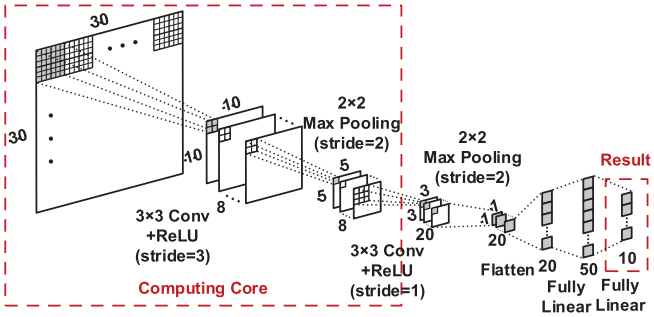


Fig. 14. Ultralightweight neural network.

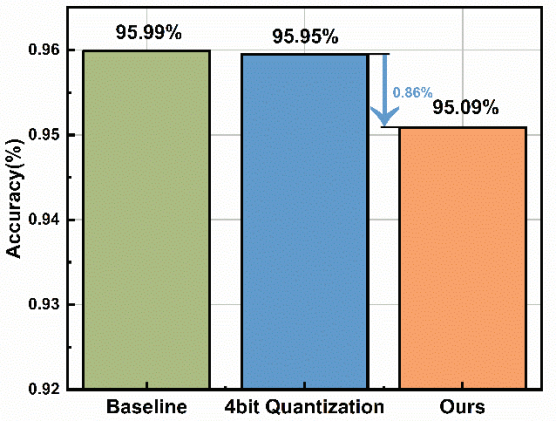


Fig. 15. Accuracy under three scenarios.

of both the input and output signals to digital values and plotted the 3-D relationship between the output, input, and weight, as shown in Fig. 12. From this plot, it can be observed that the ReLU function is activated when the weights are negative, resulting in outputs of zero. In contrast, in the region where the weights are positive, the relationship between the output, input, and weights demonstrates good linearity. Through quantitative evaluation, we calculated a root-mean-square error (RMSE) of 2.2, which corresponds to 1.2% of the full-scale range after normalization, confirming the high precision of the computation. As a visual demonstration of the image results after FE processing, the computing core was used to perform edge extraction and MP down sampling on a 120 × 120 resolution image. The computed PWM signals were proportionally converted into grayscale values and subsequently transformed into a 20 × 20 grayscale image, from which objects in the image can be roughly identified by the human eye. Fig. 13 presents the original image and the feature image obtained after FE. With a single edge extraction kernel, the computing core performs two 3 × 3 convolutions and one 2 × 2 MP operation within a working cycle of 24 μs. The PWM signal width corresponding to the maximum grayscale value of 255 is set to 7 μs. The average power consumption of the computing core is 17.53 μW, while the average power consumption of the four computing cells is 3.78 μW.

We performed classification training and testing on the MNIST dataset using the ultralightweight neural network shown in Fig. 14, which consisted of two convolutional layers,

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TABLE II
COMPARISON OF ALTERNATIVE SCHEMES

	ISSCC-2021[26]	JSSC-2021[20]	JSSC-2023[14]	Sens. J-2023[11]	ISSCC-2021[6]	**This work**
Technology	65nm	180nm	180nm	180nm	22nm	**180nm**
Supply voltage	0.8-1.2V	0.5V	Analog:0.56V Digital: 0.8V	Analog:1.2V Digital: 0.8V	Pixel and Analog:2.6V Digital: 0.8V	**0.8V**
Domain	Mixed-signal	Mixed-signal	Mixed-signal	Mixed-signal	Digital	**Mixed-signal**
Feature	1st-layer CNN, Haar-like filter	1st-layer CNN,	Event reporting, Local Binary	Edge-detection	CNN inference	**Motion detection, 3×3**
		image capturing	Pattern, raw image			**convolution, ReLU, 2×2 MaxPooling**
Conv weight precision	1.5b	±3b	N.A.	N.A.	8b/16b/32b	**[-8, +8]**
Feature result	8b(conv.) 1b(RoI)	1–8b	1.5b/ 8b /10b	10b	8b/16b/32b	**4b/ PWM**
Frame rate	24–268fps	480fps	819fps (MD), 540fps (FE)	240fps	30fps	**250fps**
FE power	42–206μW	117μW	151.9μW	17.72mW	278.8mW	**87.6μW**
MD power	N.A.	N.A.	162.6μW	N.A.	N.A.	**51.7μW**
Energy efficiency[1]	0.15–3.64	FE:1.17	N.A.	N.A.	4.97	**FE: 1.5**
FoM[2]	2.5–103.9	14.8	12.5 (MD), 17.7 (FE)	721	N.A.	**14.4-24.3**

Energy efficiency[1]: TOPS/W, FoM[2]: pJ/pixel/frame (FoM in this work is estimated based on a 120 × 120 pixel array)

two MP layers, and two fully linear layers. The initial two layers were computed utilizing the computing core, where eight convolutional kernels were employed to perform convolution operations, followed by MP. The weight quantization was set to [−8, +8]. The input was converted to PWM signals through FPGA and fed into the computing core for iterative processing. The output results were then counted and transferred to the subsequent layers for image classification. After training, the software implementation with floating-point weights achieved a classification accuracy of 95.99%. When the weights of the first layer were quantized to the range [−8, +8], the software-based accuracy decreased slightly, to 95.95%. Utilizing the computing core to perform the calculations for the first two layers resulted in a classification accuracy of 95.09%, representing a marginal degradation of only 0.86% compared to the software implementation. Fig. 15 shows the results for the classification accuracy on the MNIST dataset under these three scenarios.

The average power consumption and a power breakdown of the computing core under MD and FE modes are shown in Fig. 16. The overall power consumption of MD is reduced by a factor of 3.35× compared to FE. Fig. 17 shows the

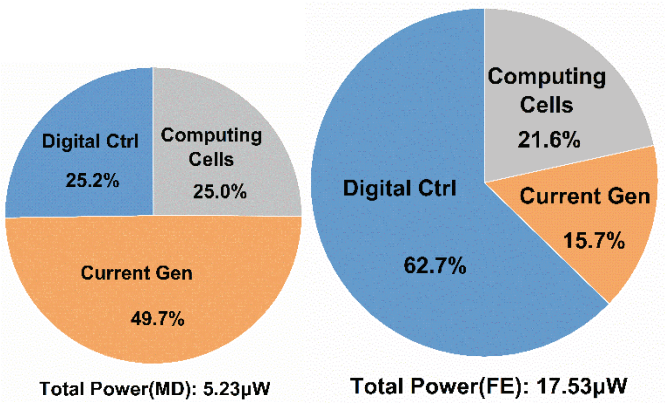


Fig. 16. Average power consumption and power breakdown of the computing core under MD and FE modes.

power consumption of the entire always-on system during operation: the system works in low-power MD mode most of the time and only wakes up the FEmode when motion is detected, thereby maintaining the average power consumption at an exceptionally low level.

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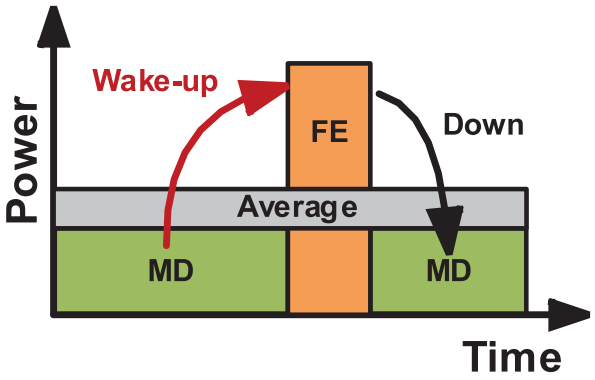


Fig. 17. Power consumption of the entire always-on system.

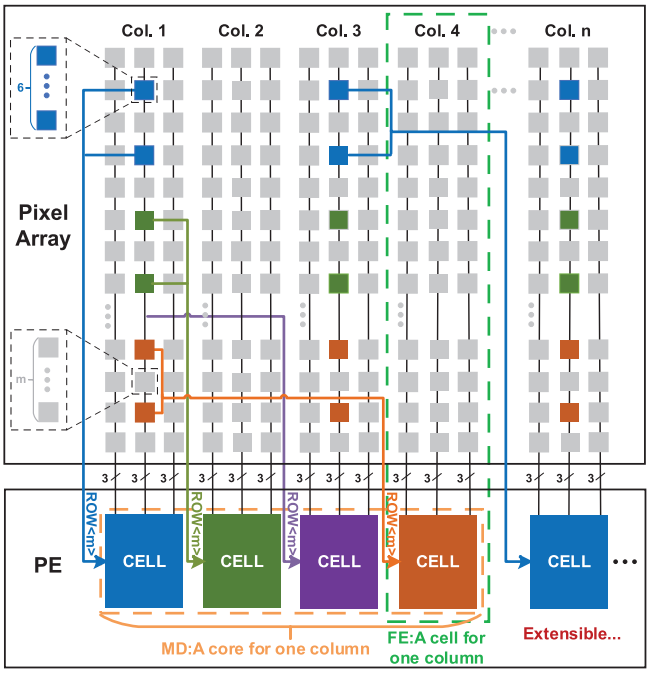


Fig. 18. Overall scheme of the proposed core for CIS.

Table II shows a performance comparison of state-of-the-art PIS technologies. Experimental results from [1] demonstrate that the power consumption of the pixel array exhibits an approximately linear relationship with frame rate. Further analysis based on the power consumption data of the pixel array provided in [20] reveals that a 128 × 128 pixel array operating at 250 fps consumes 34.7 μW, including the power consumption of the digital control module. Assuming identical pixel architecture, this study employs a 120 × 120 pixel array operating at 250 fps as the input for general-purpose computing cores. Despite the slightly lower resolution, the power consumption was conservatively estimated at 34.7 μW as the estimation basis (the actual consumption is expected to be lower). Under this configuration, ten computing cores were deployed for parallel FE operation. Based on the extrapolated power consumption of the pixel array and experimentally measured single-core power data, the total estimated power consumption in the FE mode is 87.6 μW. In the MD mode, a central 63 × 63 pixel region serves as the always-on detection window, with ten computing cores performing parallel MD operations at 250 fps, yielding a total estimated power consumption of 51.7 μW. It should be noted that the PWM presented in the feature result can be quantized to the corresponding effective number of bits according to specific application requirements. The comparative analysis demonstrates that the mixed-signal domain approach is more suitable for always-on edge devices than digital near-sensor alternatives. Furthermore, the proposed MD-FE coordination mechanism enables the designed general-purpose computing core to achieve lower system power consumption compared to existing PIS solutions, making it particularly suitable for edge devices demanding always-on operation and long-term autonomy under stringent power constraints.

V. Conclusion

This article proposes, for the first time, a general-purpose computing core integrated into the in-column architecture of CISs for PIS. It incorporates a mechanism for MD to wake up FE and utilizes PWM signals for processing. By utilizing the real-time PWM pixels output by the CIS, the computing core can perform MD or awakened FE processing on images through mixed-signal computing, without requiring extra frame memory. The voltage of both the analog and digital parts of this computing core is 0.8 V, and the core area is 720 × 272 μm. The test computing frame rate for MD and FE was 250 fps, and the total average power consumption values of the computing core in the MD and FE modes were 5.23 and 17.53 μW, respectively. The classification accuracy on the MNIST dataset was only 0.86% lower than the ideal case. Furthermore, the computing core exhibited significant scalability, enabling compatibility with PWM image sensors of varying scales and types. By flexibly deploying a corresponding number of computing cores based on the scale of the pixel array, the system can efficiently support the application requirements of multimode always-on intelligent vision sensors.

VI. Discussion

Inspired by [30], we designed a comprehensive scheme for applying the computing core to a CIS, as shown in Fig. 18. For MD, a rolling shutter exposure scheme is adopted, where each scan captures one row. A single core is utilized to sample from a full active pixel array, in which 8-pixel clusters are selected per n columns, and every two clusters are processed within one cell to complete FD. Each cluster consists of six pixels, allowing skipped pixels to be enabled and binned together to minimize blind spots. As the exposure continues, the sampled pixel information for MD is horizontally mapped to column computing cells. This mapping scheme reutilizes existing hardware resources for both frame data storage and computing. Furthermore, in addition to the uniform sampling distribution shown in the figure, the sampling points can also be irregularly distributed based on the actual application requirements, to adapt to moving targets of different shapes and varying probability distributions. If motion is detected in

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the MD mode, the FE function is activated to process the next frame. For FE, a rolling shutter exposure scheme that simultaneously scans three rows is employed, with one cell assigned to every three columns to perform MAC in parallel. FE processes all pixels. This MD-triggered FE coordination mechanism ensures the monitoring of target objects under ultralow power consumption conditions.

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